

PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

1. FUNCTIONAL REQUIREMENTS CONFIGURATION (FR):

- * System User Parameter Acceptance: The platform must cleanly capture precisely seven numeric environmental parameters (Nitrogen, Phosphorus, Potassium, Temperature, Humidity, Soil pH, and Rainfall Metric values).
- * Real-time Processing Engine Integration: The backend micro-service framework must smoothly ingest client inputs, pass them into the loaded predictive model binary (crop_model.pkl), and instantly evaluate the target array vector matrix.
- * Dynamic Visual Message Matrix Output: The client template view must output a clear, stylized response block displaying the optimal recommended agriculture classification without resetting browser state parameters entirely.

2. NON-FUNCTIONAL REQUIREMENTS SPECIFICATION (NFR):

- * Real-Time Analytical Response Execution: Total operational processing turnaround speed from backend matrix multiplication to client screen output should stay strictly below 2.0 seconds.
- * High Algorithmic Classification Score: The predictive classifier engine algorithm must achieve over 90% optimization match accuracy given consistent analytical laboratory sample ranges.
- * Clean Responsive Visual Interface: UI grid elements layout structures must accurately dynamic-resize onto multiple client viewing frames (Tablets, Laptops, Mobile view addresses).